Alex Layton

Associate Principal Software Engineer | Systems Architecture & Connected Products
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Profile

Associate Principal Software Engineer with over 13 years' experience across mobile, cloud and embedded systems. At Dyson, takes connected-product capabilities from proof of concept to architecture, reusable components and adoption by delivery teams, across smart-home commissioning, device connectivity, product visualisation and developer platforms. Pairs hands-on engineering with technical leadership across app, cloud and embedded teams. Currently focused on adopting AI in software engineering: using and building agentic tools and workflows that take on routine development work and shorten the path from idea to working prototype.

Technical skills

Architecture & Leadership: Systems architecture, Cross-team technical leadership, Proof-of-concept development, Reusable platforms, Mentoring
Languages (most proficient first): Swift, Python, Rust, JavaScript/TypeScript, Go, Kotlin/Java, C/C++
AI & Agentic Tools: LLM integration, Agentic workflows, MCP
Connected Systems: AWS IoT, MQTT, BLE, Matter/Thread, Wi-Fi provisioning, HomeKit
Cloud & Delivery: AWS (Lambda, ECS, EC2, S3, DynamoDB), Docker, CI/CD, Bitrise, Jenkins, GitOps, gRPC, PostgreSQL
Client Engineering: Swift Packages, SwiftUI, React

Experience

Associate Principal Software Engineer, Dyson — Bristol, UK *Oct 2021 – Present*

- **Defined the Matter provisioning architecture used in Dyson's De-NOx purifiers.** Designed the MQTT provisioning specification and aligned app, cloud and embedded teams around it. Validated the approach with a C++ accessory and an iOS app, then delivered commissioning libraries, including the enhanced commissioning payload generation used in MyDyson for Android.
- **Migrated the team's CI/CD from Jenkins to Bitrise.** Validated monorepo and trunk-based workflows with Bitrise engineers. Automated deployment of changed apps to cut time spent distributing builds by hand, and built Rust tooling to integrate internal services.
- **Architected and built OneHub, an internal app distribution platform.** Designed for new product development and feature trials, it combines modular Swift Packages, SSO, permission-based access and ephemeral downloads with GitOps-driven releases and beta distribution. Rollout is in progress.
- **Co-developed Dyson3D, a reusable schema-driven rendering framework.** Shipped in MyDyson for OnTrac headphone customisation, with configurable colourways, attachments and animations. Added dynamic snapshot generation to reduce bundled image assets.
- **Validated robot-map integration for Spot+Scrub AI with an external partner.** Worked with the partner to integrate Dyson3D for interactive 3D map visualisation in MyDyson.
- **Developed audio-source selection with embedded engineers.** Supported app delivery teams with post-launch integration for Dyson Zone and OnTrac headphones.

Lead Software Engineer, Dyson — Bristol, UK *Apr 2020 – Oct 2021*

- **Engineering lead for the Air Quality Backpack companion app.** Led a small team on the King's College London collaboration, delivering device communications and background synchronisation so air-quality data uploaded reliably.
- **Built reusable Swift Packages for connected-product prototypes.** Let engineers build features against production protocols and endpoints instead of rebuilding integrations.
- Mentored undergraduates on the Dyson Institute university programme.

Senior Software Engineer, Dyson — Malmesbury / Bristol, UK *Jan 2018 – Apr 2020*

- **Helped validate the migration from RabbitMQ to AWS IoT.** Worked with Amazon engineers on a proof of concept, then delivered dependency-free MQTT clients for iOS and Android to app delivery teams.

- **Built a HomeKit prototype that secured business support for smart-home development.** Brought existing connected products to Apple Home, then contributed to a production-intent MFi implementation before the team moved to Matter.
- Built interactive 2D mapping concepts with robotics teams, supporting mapping and zoning for the Dyson 360 Heurist and 360 Vis Nav.

Software Engineer, Dyson — Malmesbury / Bristol, UK *Jan 2017 – Jan 2018*

- **Helped replace SoftAP onboarding with BLE and Wi-Fi provisioning.** Designed, implemented and tested the connection journey with embedded teams, then supported its rollout in DysonLink (now MyDyson).
- Built iOS and Android apps with embedded teams to validate Dyson's first BLE-only connected machines, then helped delivery teams integrate them.

Mobile Developer, Ventutec — UK *Oct 2014 – Dec 2016*

- Developed and maintained commercial iOS and Android apps and scoped new applications with clients. Prototyped a BLE-based IoT app and cloud platform.

Graduate Analyst Programmer, Graffica — UK *Sept 2013 – Oct 2014*

- Developed airspace-planning software used across Europe, and automated the software release process for the Airspace Management team.

Education

University of Birmingham *2010 – 2013*

- BSc Computer Science — First Class with Honours

Selected projects

Harness — Rust coding agent (personal / open source)
https://github.com/alexlayton/harness
A Rust coding agent with streaming multi-provider LLM integration, tool execution and bounded subagents. Supports resumable append-only sessions, context compaction, indexed code search and MCP tools, through terminal, headless and ACP editor interfaces.